AI in Finance: Credit Risk Classification

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Context of Problem: Why Credit Risk Modelling Matters

- Credit scoring underpins financial inclusion, determining who can access loans, mortgages, or business credit.
- Biased or erroneous credit classification can exclude individuals or artificially inflate the risk of default to lenders.
- If left unregulated, AI models can perpetuate historical discrimination or data bias that exacerbate inequality.
- Transparency, human-in-the-loop processes, and algorithmic accountability are mandated by regulators now, for example through GDPR Article 22 and the proposed EU AI Act.
- The challenge: building models that are accurate, fair, and explainable-not just statistically powerful.

The Role of AI in Finance

- AI is changing the way financial services organizations evaluate, grant and administer credit.
- Algorithms now drive decisions made by human analysts, from loan approvals to interest rate assignment.
- AI-lead credit scoring improves efficiency while raising ethical and transparency concerns in high-stakes decision-making.
- This project evaluates the classification of credit risk into Low, Average and High categories, using Logistic Regression, K-Nearest Neighbours, XGBoost and Neural Networks.

Research Gap

- Traditional models, such as Logistic Regression, are relatively simple and interpretable but cannot capture non-linear financial behavior.
- Advanced models such as XGBoost and Neural Networks have high predictive accuracy but are often treated as "black boxes".
- KNN has intuitive classification but suffers with large, high-dimensional datasets.
- This creates a trade-off between performance and interpretability-crucial for responsible AI in finance.
- Our study aims to fill this gap by comparing four algorithms in a similar preprocessing and evaluation set-up.

Project Aim and Objective

- Develop and train four models using machine learning for credit score classification. By implementing a leak-free data preprocessing pipeline, perform encoding, validation and tuning.

- Compare algorithm performance using:

- Confusion Matrix: class-level accuracy and misclassification trends.
- ROC Curve: Discriminative ability across thresholds.
- SHAP Values: explainable feature influence for fairness.

- Assess the best performance model and most aligned with standards on ethical and interpretable AI.

Financial Data

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Preprocessing Pipeline

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Machine Learning Models

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Model Evaluation

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Ethical & Fair Decision-Making

Hypothesis:

Integrating explainability, into complex models can deliver both performance and interpretability.

Expected results:

- XGBoost: Best overall accuracy with balanced interpretability.
- Logistic Regression: Transparent but limited performance.
- KNN: Moderate interpretability, weaker scalability.
- Neural Network: High learning capability, low explainability.

Broader impact: Showing that ethical, transparent, and high-performing AI systems can be delivered for real-world finance

Algorithms Used

Gizem (XGBoost)

I chose XGBoost because it's the strongest algorithm for tabular financial data, as it captures non-linear patterns. It is also great with mixed feature types and works well with SHAP, making it accurate and explainable.

XGBoost builds decision trees with each tree correcting the errors of the previous one through gradient boosting.

Regularisation and random subsampling prevent overfitting. Final output is a probability distribution.

Paaralathan (Logistic Regression)

I used Logistic Regression, a method of supervised learning that can categorise data as low, average, or high credit risk.

I picked Logistic Regression because it is easy to understand, interpretable, and produces clear probability results. It is also commonly used in financial risk analysis to show which factors have the most impact on the result.

Logistic regression is one of the most often used methods for classification. It calculates the probability that a given input belongs to a specific class.

Arafat (KNN)

KNN was chosen for its simplicity and interpretability as the concept is straightforward to understand making it a good start for classification problems.

It works by finding the k nearest neighbours which represent borrowers with very similar financial behaviour to a given input, and makes a prediction based on majority class, hence the outcomes are highly relevant for predicting a new applicants risk.

Saihaan (Neural Network)

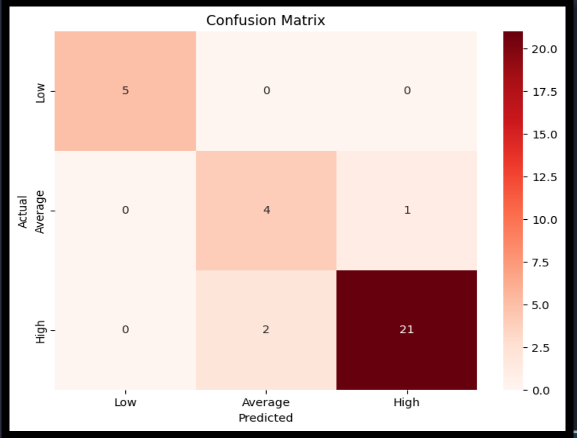
The algorithm I chose is a multilayer perceptron (MLP) neural network. This is an artificial neural network that is made up of connected neurons. It learns patterns by adjusting internal weights through training, which allows it to model non-linear relationships between things like age, Income, marital status etc.

I chose a MLP because of its ability to capture complex relationships, its compatibility with tools like SHAP that help to explain the model choices and its flexibility to change the number of neurons and layers to improve accuracy.

Credit Risk Prediction - Confusion Matrix

Figure 1 (Paaralathan)

Figure 2 (Arafat)



Logistic Regression, the basic linear model meaning it can only draw a straight light whereas complex models can curve (non-linear), which is why logistic regression performed much worse than the advanced models. KNN performed better with 91.0% accuracy because of its ability to handle non-linear complexity, these are . The difference between low, average, and high credit scores is very complex. XGBoost and Neural Networks, can be more accurately detected when it comes to predicting high-risk (low credit score) customers, these are advanced non-linear algorithms.

The diagonal of the Confusion Matrix shows accurate predictions, while the opposite side displays errors. In credit lending, the most serious mistake is predicting a risky borrower as low-risk (a False Negative). To reduce financial loss, we can focus on Recall in the “Low Credit Score” class, which shows how well risky customers are detected so we can avoid them. While Neural Networks achieved high accuracy, its lack of transparency makes loan decisions harder to justify under GDPR.

Logistic Regression (Figure 1)	KNN (Figure 2)
1. Correct predictions: 3 Low, 6 Average, 23 High 2. Errors: 1 Average predicted as High. 3. Shows strong diagonal dominance, which means nearly all predictions were correct. 4. Confusions in the “Average” group but it has perfectly identified all low credit customers with a recall of 1.00.	1. Correct predictions: 5 Low, 4 Average, 21 High 2. Errors: 1 Average as High, 2 High as Average. 3. Made mistakes because some groups had similar values, making it hard to tell apart. It’s also very sensitive to how the numbers are scaled. 4. Performs well overall but it's less stable than Logistic Regression or XGBoost.

XGBoost (Figure 3)	Neural Network (Figure 4)
1. Correct predictions: 15 Low, 36 Average, 113 High 2. Errors: No errors it’s a perfect diagonal. 3. It has perfect classification and excellent handling of imbalance. 4. Best at identifying risky borrowers, with accuracy at 100% and recall at 1.00.	1. Correct predictions: 5 Low, 5 Average, 21 High 2. Errors: 2 High predicted as Average. 3. High overall accuracy 94% but slightly overfit the data, it was too confident and made small mistakes. 4. XGBoost and NN are highly accurate but less transparent, fairness problems in decision-making.

Figure 3 (Gizem)

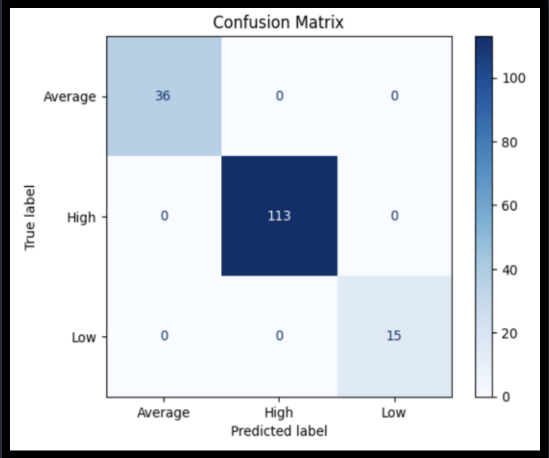
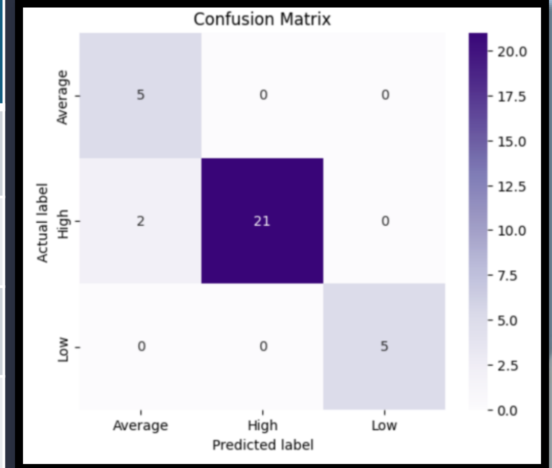


Figure 4 (Saihaan)



Model	Overall Correctness (Accuracy)	Balanced Performance (F1-Score)	High-Risk Detection (Low Credit Recall)	Analysis
Logistic Regression	97.0%	0.97	1.00	With a 97% accuracy and a F1-score of 0.97, it performs very strong and correctly identified all high-risk customers.
K-Nearest Neighbors (KNN)	91.0%	0.91	0.80	91% accuracy and an F1-score of 0.91 but missed some high-risk cases (recall 0.80), making it less dependable for risk detection.
XGBoost	100%	1.00	1.00	100% on everything, It handles complex patterns very well with no errors, making it the most reliable model for detecting high-risk.
Neural Network (NN)	94%	0.94	1.00	94% accuracy and F1-score of 0.94, strong overall performance. Detected all high-risk customers but some small mistakes with Average group.

Credit Risk Prediction - ROC Curve

The ROC curve is a representation of model performance across all thresholds. It is drawn by calculating the true positive rate (TPR) and the false positive rate (FPR). The AUC (area under curve) represents the probability that a randomly chosen positive is ranked higher than a randomly chosen negative. An AUC of 1 indicates a perfect classification.

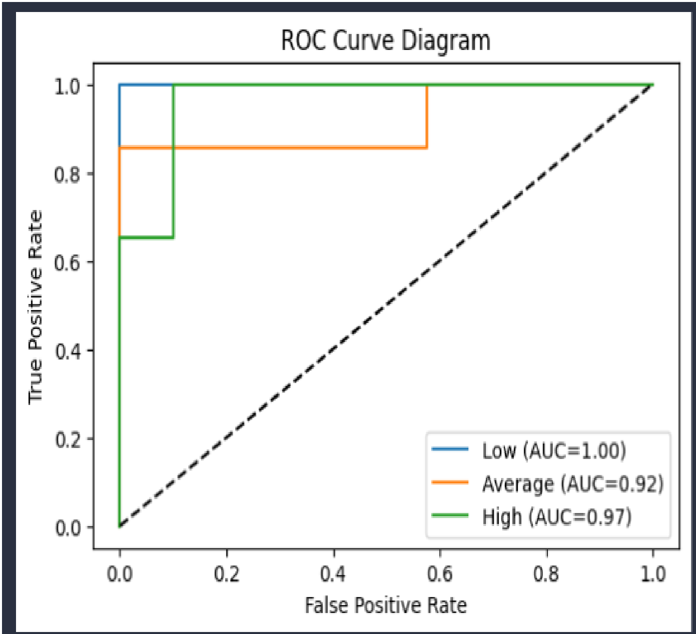
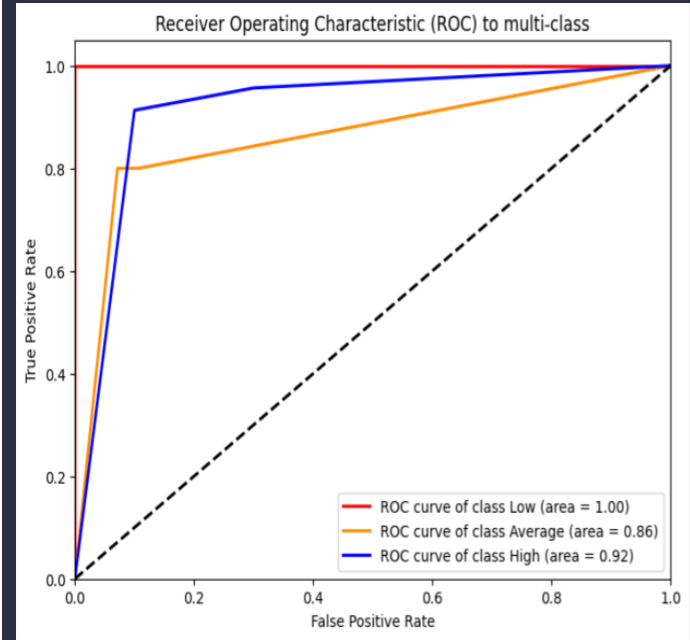


Figure 1 (Paaralathan)

Figure 2 (Arafat)



Logistic Regression (Figure 1)	KNN (Figure 2)
1. Class Low has an AUC of 1, indicating a perfect classification. The model correctly identifies all cases of Low credit with no false positives. 2. Class High has an AUC of 0.97, which is an excellent classification. This means that the model correctly identifies all cases of High credit with 97% accuracy. 3. Class Average has an AUC of 0.92, which is very good but lower than the other classes. On average, the model can correctly distinguish each class from the other 92% of the time.	1. Class Low has an AUC of 1, which is a perfect classification, correctly identifying all cases of low credit perfectly. 2. Class High has an AUC of 0.92, which is lower than all the other models. However, this is still a high classification, which means the model correctly identifies all cases of high credit with 92% accuracy. 3. The Class Average has an AUC of 0.86, this indicates that the model is performing well, but not flawless, as there will be some overlap between groups.
XGBoost (Figure 3)	**Neural Network (Figure 4)**
1. Class Low has an AUC of 1, indicating perfect classification, correctly identifying all cases of Low credit with no errors. 2. Class High has an AUC of 1, which is also a perfect classification, the model correctly identifies all cases of high credit with no false positives. 3. Class Average has an AUC of 1, indicating a perfect classification. This means that the model is flawless and correctly identifies all cases with no overlaps or errors.	1. Class Low has an AUC of 1, indicating a perfect classification, where all cases of low credit are identified with no errors/false positives. 2. Class High has an AUC of 0.98, indicating an excellent classification, this model almost perfectly correctly identifies all cases of high credit. 3. Class Average has an AUC of 0.96, which is still very high but slightly lower than class low and class high. This class overlaps with both other classes, so the model finds it harder to separate.

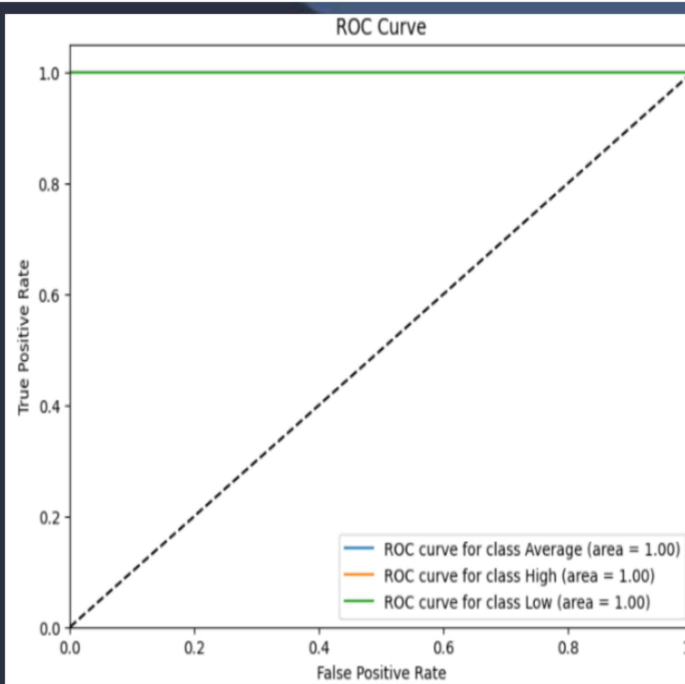
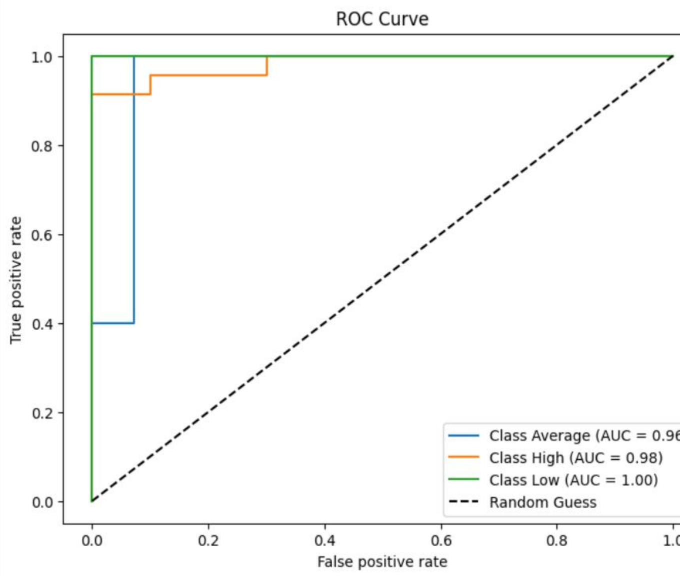


Figure 3 (Gizem)

Figure 4 (Saihaan)



Credit Risk Prediction - SHAP

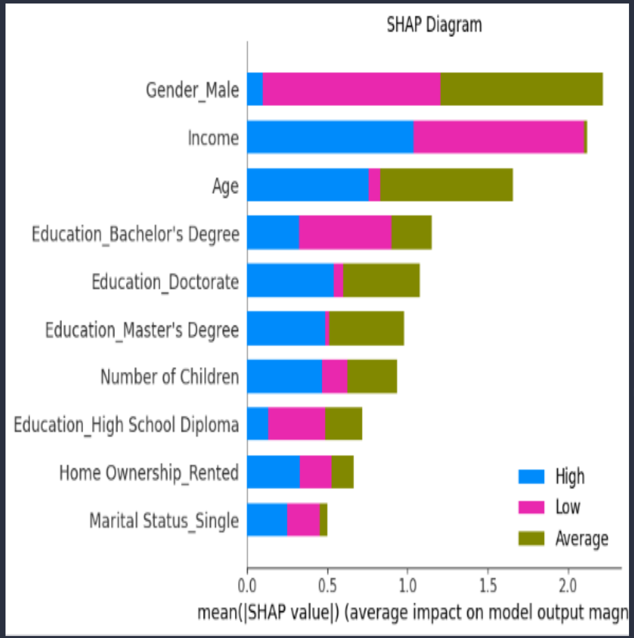


Figure 1 (Paaralathan)

SHAP is used to explain the output of any machine learning model. For credit risk, SHAP helps us understand why a person is high risk or low risk. The following SHAP plots show the most important features for each model and the impact it has on the models output.

Logistic Regression (Figure 1)	KNN (Figure 2)
1. Most important features : Gender_male , Income and Age 2. Impact: Gender_male is the most important feature, the model relies on it the most to make credit risk predictions. 3. Interpretation: Gender_male produces low risk along with average risk. Income has an even split between high risk, and low risk which means very low and very high incomes are risky. Age shows how it's often linked with high and average credit risk and a little low risk. 4. Spread :The spread for logistic regression compared to the others is well balanced	1. Most Important features : Income , Age and Number of Children 2. Impact: Income is the most important feature, the model relies on it the most to make credit risk predictions. 3. Interpretation: Income gets the most high risks. The low and average risks are similar to that of Age. Number of children gives high average and medium low and little high. 4. Spread :The spread for KNN is not as balanced as logistic regression as there is a clearer dependency on the top 3 which is income, age and number of children.

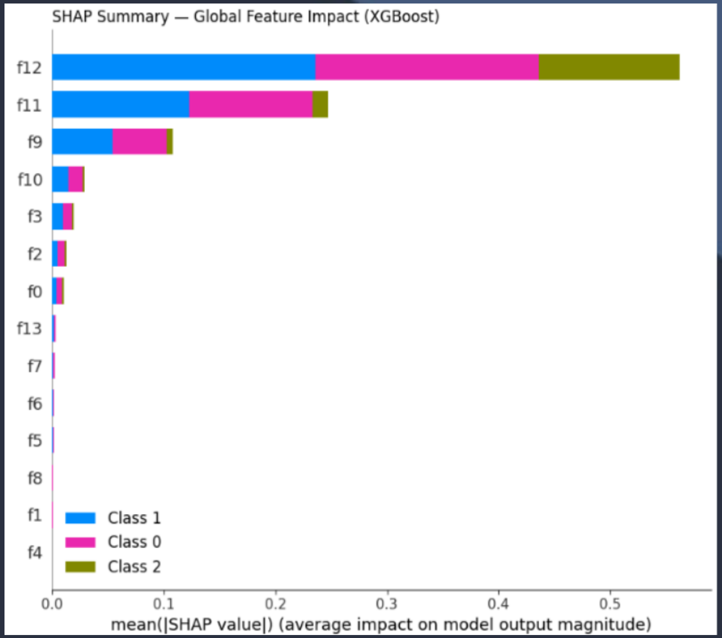
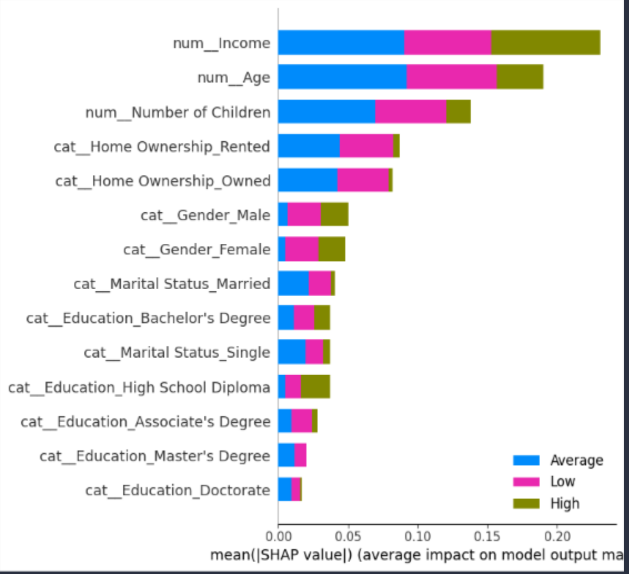


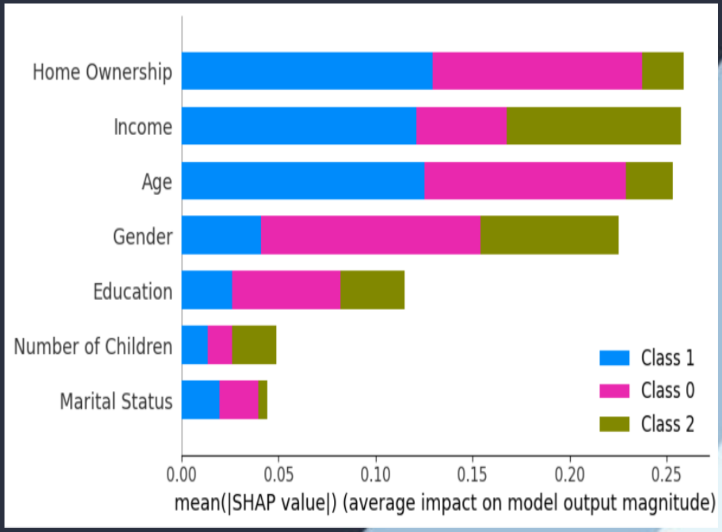
Figure 3 (Gizem)

Figure 2 (Arafat)



XGBoost (Figure 3)	Neural Network (Figure 4)
1. Most important features: f12 (Home Ownership) , f11 (Income), f9 (Education Level) 2. Impact: F12 is the most important feature the model relies on it the most to make credit risk predictions 3. Interpretation :f12 has the strongest impact influencing predictions for Class 1 and Class 0 and then Class 2. f11 is the next most important feature affecting Class 1 and Class 0 the most. f9 influences Class 1 and Class 0 predictions, but to a lesser amount than f12 and f11. 4. Spread: The spread for XGBoost is out of proportion with f12 being the clear top feature and after f11 and f9 the rest are negligible.	1. Most important features : Home ownership, Income and Age 2. Impact : Home ownership is te most important feature, the model relies on it the most to make credit risk predictions 3. Interpretation: Home ownership are split across class 1 and class 0.Income with a large class 1 and class 2.Age class 1 and class 0. 4. Spread: One of the better spreads almost all balanced with 4 out of 7 nearly all influential.

Figure 4 (Saihaan)



Conclusion

Overall Performance and Ranking

1. XGBoost - Best Overall: Highest accuracy and perfect ROC AUC across all classes. Combines predictive strength with SHAP-based explainability, thus directly solving the trade-off between accuracy and transparency.

2. Logistic Regression - Reliable & Transparent: Moderate accuracy but fully interpretable, perfect for risk-averse financial environments.

3. Neural Network - High Learning Power, Low Explainability: Performed AUC well, but less interpretable and at risk of overfitting, hence less trustworthy for real-world finance applications.

4. KNN - Simplistic & Inconsistent: Lowest recall for high-risk clients; performance sensitive to feature scaling, which reduces stability.

Ethics and Professional Reflection

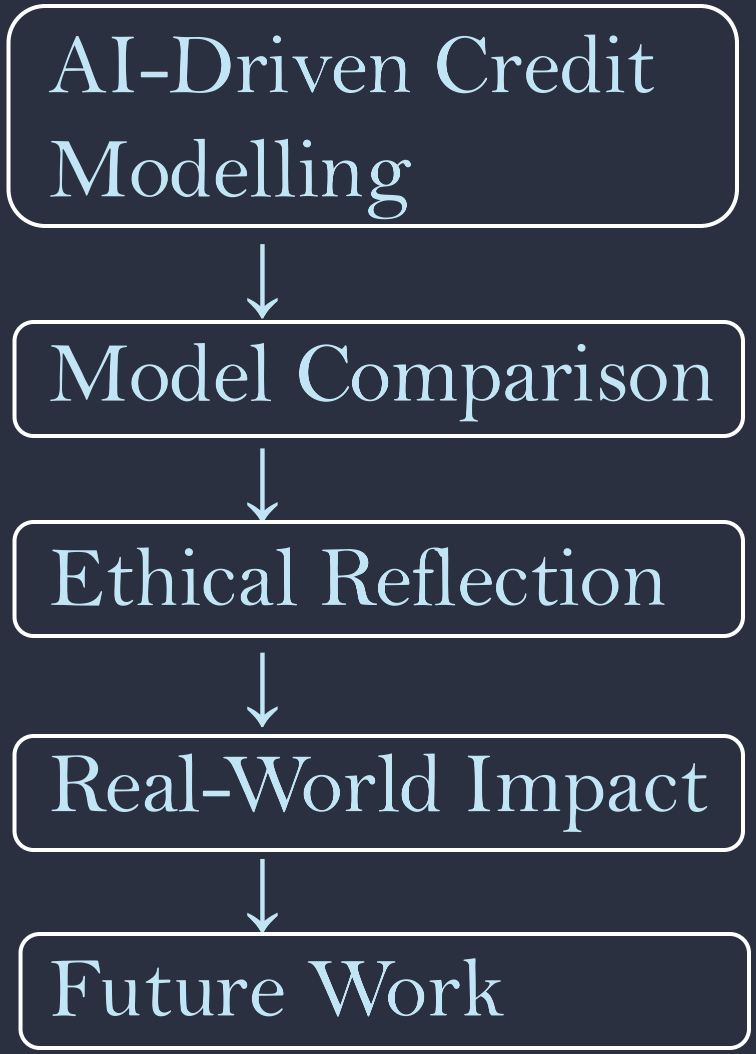
- Fairness & Transparency: XGBoost with SHAP ensures explanations to automated decisions under the standards of Article 22 of GDPR and the EU AI Act.
- Accountability: SHAP visualization explains feature impact, reducing biases in credit decisions.
- Trust in AI: Transparent reasoning builds user and regulatory confidence.
- Professional Responsibility: Models should be documented, version-controlled and bias-tested to ensure financial accountability.

Future Directions

- Expand testing for more applicability on larger and practical datasets.
- Implement fairness-aware metrics, including demographic parity and equal opportunity.
- Incorporate local SHAP analysis for individual borrower explanations.
- Develop hybrid interpretable models that combine logistic transparency with ensemble accuracy.

Real world impact

- It allows for fair approval of loans and reduces the risk of default to lending institutions.
- Enhances financial inclusion, especially within the underbanked communities.
- Supports responsible data-driven lending ecosystems in line with sustainable finance principles.



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Link to our presentation: https://youtu.be/fBinfF3Fnhk